abstract-audit: TMLR figure and table sources

Main text

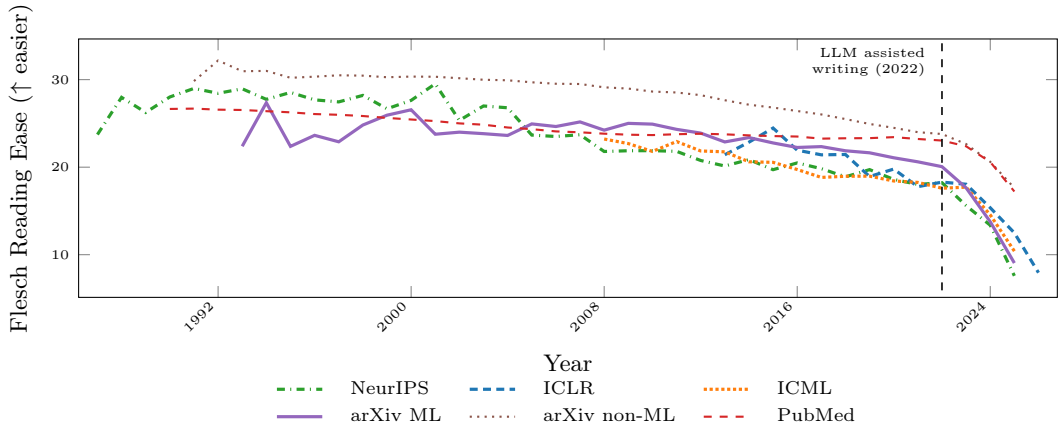


Figure 1: **Flesch Reading Ease across all six corpora (NeurIPS, ICLR, ICML, arXiv ML, arXiv non-ML, and PubMed), 1987–2026.** Higher values indicate easier reading (↑). The three conferences and arXiv ML all decline, and by 2025 each sits below both the arXiv non-ML and the PubMed baseline.

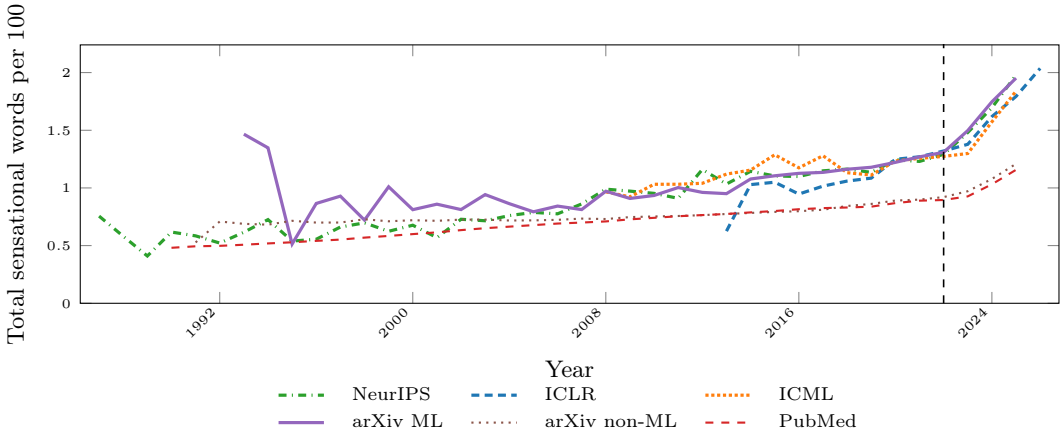


Figure 2: **Total sensational language per 100 abstract words, 1987–2026.** The rate rises across all three conferences and arXiv ML. The arXiv non-ML baseline rises far more slowly, and the divergence between ML and non-ML venues accelerates after 2022. The breakdown by category appears in Appendix **??**.

Family	What it measures	Section	Direction over time
Classical readability, 15 formulas	Surface difficulty of the prose, from sentence length and word length	??	Harder to read at every venue
Hohmann writing style suite, extended to 12 metrics	Syntactic structure, including parse depth, passive rate and noun phrase density	??	Largely parallel to the non-ML control
Sensational language rate, part of the style suite	Rhetorical stance, counted as promotional words per 100 words	??	Rising at every venue
Acronym density and reuse	Vocabulary a reader must already hold, and how often each term recurs	??	Denser, with most terms rarely reused
LLM as judge, 6 models	What an automated reviewer reports about readability	??	Rising after 2022 at NeurIPS and ICML, against the classical scores
Citation counts	What the field rewarded, within venue-year cohorts	??	More readable papers cited slightly more

Table 1: The five metric families, what each one measures, and the direction it moves at NeurIPS, ICLR and ICML.

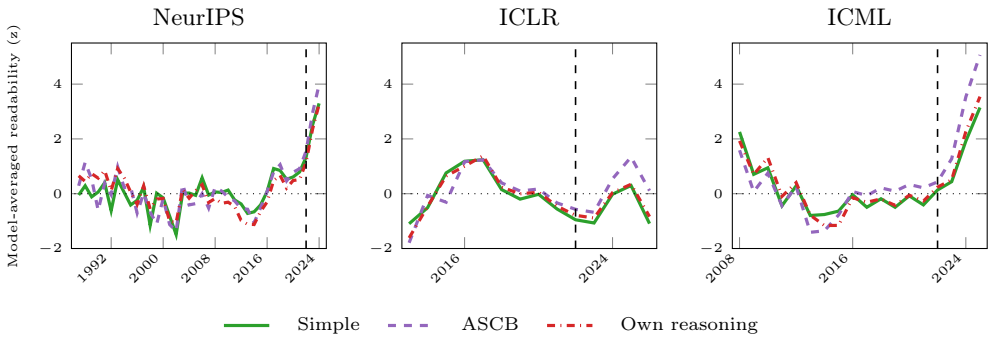


Figure 3: **LLM as judge readability across the three conferences.** Each curve is the mean z-score across all six open-weight models for one prompt variant, standardized to each model's own baseline through 2022 within that venue. NeurIPS and ICML show the same pattern. Scores rise sharply after 2022, by roughly three to five standard deviations, directly contradicting the classical readability decline in Figure 1. ICLR shows no comparable rise, and its post-2022 movement does not settle into a consistent trend across the three prompts, so we draw no conclusion from that series and leave it to further investigation.

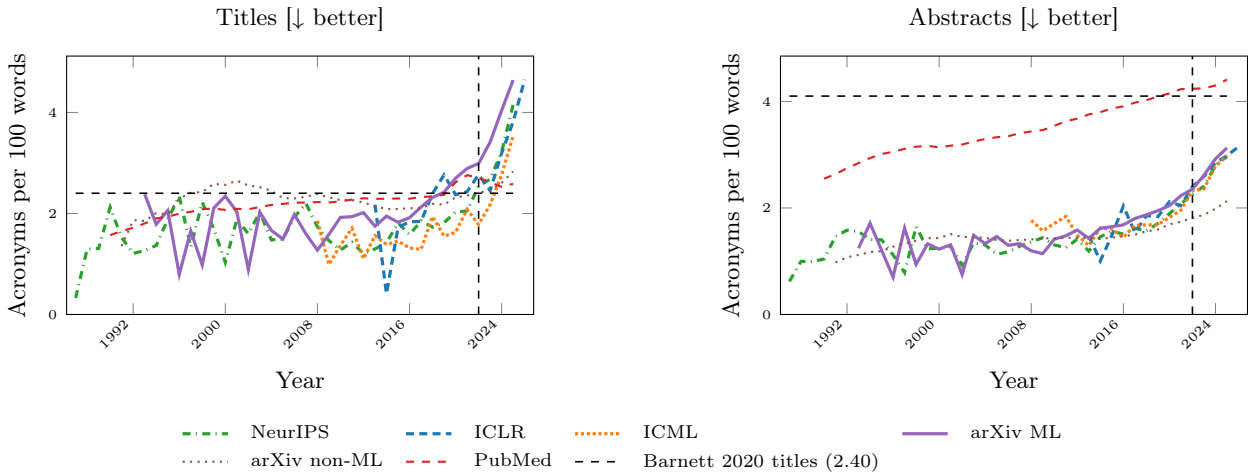


Figure 4: **Acronym density in titles (left) and abstracts (right), 1987 to 2026. ?**] report acronym density for titles and for abstracts separately, and their 2019 biomedical values differ between the two. The dashed black horizontal line marks the applicable one in each panel, 2.40 per 100 words for titles and 4.10 for abstracts. ML title density has risen above the biomedical title value, while ML abstract density remains below the biomedical abstract value.

ID	Leading terms	Papers	Flesch	Med. cit.
T1	diffusion, image, diffusion models, generation	3,551	9.7	43
T2	reasoning, llms, language, language models	3,765	11.1	52
T3	task, tasks, domain, data, performance, knowledge	5,734	13.4	35
T4	3d, video, object, scene, view, image, objects	2,995	13.7	32
T5	inference, latent, bayesian, data, variational	5,051	13.8	28
T6	reinforcement, reinforcement learning, policy, rl	5,172	14.0	31
T7	data, machine, machine learning, training	5,889	14.8	26
T8	language, llms, language models, large language	5,036	14.8	52
T9	gradient, optimization, convergence, convex	3,135	15.2	28
T10	supervised, self, supervised learning	2,450	16.9	38
T11	neural, neurons, network, brain, networks, visual	2,247	18.1	16
T12	graph, graphs, gnns, graph neural, node, networks	2,506	18.2	32
T13	networks, neural, neural networks, network, deep	5,680	19.4	38
T14	algorithm, problem, data, algorithms, linear	6,697	20.8	23
T15	regret, algorithm, online, bandit, algorithms	2,720	21.9	18

Table 2: The topics induced from the cleaned titles and abstracts of all conference papers, ordered from least to most readable, with the leading terms of each cluster centroid. Median citations are computed on the papers with citation data. Readability spans 12.1 Flesch points across topics and correlates negatively with citations across them ($r = -0.59$), the opposite sign to the within-topic association.

Appendix

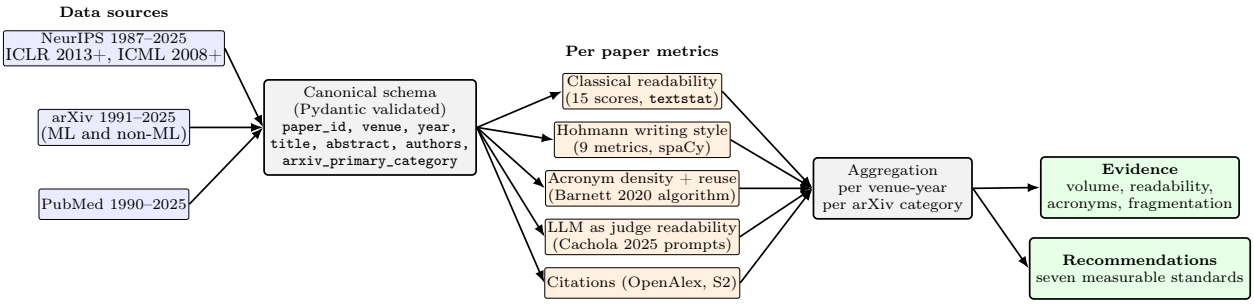


Figure 5: **End to end workflow.** Three data sources feed a canonical Pydantic-validated schema. Five per-paper metric families are computed on abstracts, and acronym density also on titles, then aggregated per venue-year and per arXiv primary category. The outputs feed the empirical evidence (Sections **??** through **??**) and the recommendations (Section **??**). Pipeline details appear in Appendix **??**.

Source	Coverage	Papers
arXiv (all categories)	1991–2025	2,823,623
NeurIPS	1987–2025	30,595
ICLR	2013–2026	16,807
ICML	2008–2025	15,226
PubMed	1990–2025	24,306,644

Table 3: Per source paper counts and coverage at the 2026-04 snapshot. The arXiv 2025 row covers approximately three quarters of the calendar year. The NLM baseline additionally carries 156,564 ahead-of-print PubMed records dated 2026 and 2027. An ahead-of-print date is not a publication year, so these are excluded from the PubMed count above and from every year-indexed analysis and figure, and from the pooled acronym reuse distribution of Table 10.

Element	Before	After
Inline mathematics	`coresets ($\texttt{FWC}$)`	`coresets ( x )`
Display mathematics	`regret is $O(\sqrt{T})$`	`regret is x`
Mathematics spliced inside a word	`$\underline{Co}$llabo-$\underline{ra}$tive`	`Collaborative`
Text-mode commands	`the \emph{\textbf{same}} setting`	`the same setting`
URLs	`available at https://-github.com/ex/repo`	`available at`
Identifiers	`see arXiv:2401.00001`	`see`
Ampersands	`speed & accuracy`	`speed and accuracy`
Non-ASCII characters	`the café corpus`	`the caf corpus`
Section headings	`OBJECTIVE: We study scaling.`	`We study scaling.`

Table 4: Text normalisation rules, with the effect of each on a worked example. Applied for the sensitivity analysis in Table 5. The reported metrics are computed on the abstract as published.

Normalisation	Venue	First year	Last year	Change
None	NeurIPS 1987–2025	23.73	7.59	−16.14
LaTeX only	NeurIPS 1987–2025	23.74	7.92	−15.82
LaTeX and URLs	NeurIPS 1987–2025	23.74	8.44	−15.31
None	ICLR 2013–2026	21.42	7.94	−13.48
LaTeX and URLs	ICLR 2013–2026	21.52	8.74	−12.78
None	ICML 2008–2025	23.22	10.41	−12.81
LaTeX and URLs	ICML 2008–2025	23.22	11.23	−11.99

Table 5: Flesch Reading Ease at each venue's first and last year under three text normalisations. Markup accounts for 5 to 6 percent of each measured decline, and URLs for more of it than LaTeX.

Simple Prompt

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. Assume the reader is an adult. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

ASCB Guidelines Prompt

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. Characteristics of a highly readable text include: \n
– Know your audience, and focus and organize your information for that particular audience. \n
– Focus on the big picture. What larger problem is your work a part of? What major ideas or issues does your work address? How will your work help global understanding of some issue? \n
– Avoid jargon. If you must use a technical term, make sure to explain it, but simplify the language. \n
– Try to use metaphors or analogies to everyday experiences that people can relate to. \n
– Underscore the importance of public support for exploratory research and scientific information, and the role of this information in providing the context for effective policy making. \n \n
Assume the reader is an adult. Do not use Flesch-Kincaid or other readability formulas. Use your own judgment to rate the text. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

Own Reasoning Prompt

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. \n
Assume the reader is an adult. Do not use Flesch-Kincaid or other readability formulas. Use your own judgment to rate the text. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

Table 6: The three prompt templates we apply, reproduced verbatim from **?**] (Table 10) except that the Reason: <reasoning> output line is omitted. {SUMMARY} is replaced by the abstract text at inference.

Model	β	s.e.	95% CI	within R^2
No topic control	0.049	0.008	0.034 to 0.063	0.066
Topic fixed effects, $k = 10$	0.042	0.006	0.030 to 0.055	0.148
Topic fixed effects, $k = 15$	0.033	0.007	0.019 to 0.046	0.162
Topic fixed effects, $k = 20$	0.034	0.006	0.022 to 0.045	0.166
Topic fixed effects, $k = 30$	0.036	0.007	0.023 to 0.050	0.161

Table 7: Standardised coefficient on Flesch Reading Ease in the citation regression, with and without topic fixed effects. Venue-year fixed effects, abstract length and author count are controlled throughout, and standard errors are clustered by venue-year.

ID	Leading terms	Papers	Span	ΔFlesch	ΔHype	ΔAcronyms
T3	task, tasks, domain, data, performance, knowledge	4,957	2015–2025	−16.7	+0.56	+1.90
T12	graph, graphs, gnns, graph neural, node, networks	2,250	2018–2025	−13.2	+0.76	+2.35
T2	reasoning, llms, language, language models	2,572	2021–2025	−12.2	+0.49	+1.48
T8	language, llms, language models, large language	4,042	2020–2025	−11.3	+0.59	+1.24
T5	inference, latent, bayesian, data, variational	3,685	2015–2025	−11.1	+0.32	+0.48
T11	neural, neurons, network, brain, networks, visual	891	2016–2025	−11.1	+0.41	+0.80
T4	3d, video, object, scene, view, image, objects	2,442	2018–2025	−11.0	+1.20	+1.37
T13	networks, neural, neural networks, network, deep	5,065	2015–2025	−10.5	+0.38	+0.78
T1	diffusion, image, diffusion models, generation	2,958	2018–2025	−9.6	+0.90	+0.30
T15	regret, algorithm, online, bandit, algorithms	2,336	2015–2025	−8.8	+0.29	+0.67
T7	data, machine, machine learning, training	5,255	2015–2025	−8.4	+0.19	+1.20
T10	supervised, self, supervised learning	2,045	2016–2025	−7.8	+0.78	+0.91
T6	reinforcement, reinforcement learning, policy, rl	4,445	2016–2025	−7.7	+0.49	+1.73
T9	gradient, optimization, convergence, convex	2,760	2015–2025	−5.7	+0.30	+0.52
T14	algorithm, problem, data, algorithms, linear	4,457	2015–2025	−4.6	+0.13	+0.47

Table 8: Change in three of the five metric families within each topic, measured from the first to the last year in which that topic has at least 30 papers within the 2015 to 2025 window. Several topics have fewer than 30 papers at the start of that window. Flesch Reading Ease is in points, sensational language and acronym density in occurrences per 100 words. Paper counts are for the window, so they are lower than the full-corpus counts in Table 2, whose identifiers these match.

Weighting	Within topic	Between topic	Within share (%)
Topic mix held at 2015	−9.1	−1.8	83
Mix averaged over both years	−13.3	+2.4	122
Topic mix held at 2025	−17.6	+6.7	161
Gap to papers of the same year	−11.0	+0.1	101

Table 9: The fall of 10.9 points in mean Flesch Reading Ease across the three conferences from 2015 to 2025, split into a part within topics (topics becoming less readable) and a part between topics (the field moving between topics). Each row sums to the total. The first three rows value each topic at its own mean score and differ only in which year's topic mix weights the within-topic change. The last row values each topic by its gap to papers of the same year, estimated over every year of the window, which avoids topic means that rest on a few papers in 2015. A positive between-topic entry means the change in topic mix raised readability.

Venue	Unique acronyms	1	2–9	10–99	100–999	1000+
NeurIPS	15,693	**31.2%**	58.3%	9.5%	1.0%	0.0%
ICLR	11,097	**27.0%**	63.2%	8.8%	0.9%	0.0%
ICML	8,218	**26.9%**	62.8%	9.6%	0.7%	0.0%
arXiv ML	115,229	**28.9%**	59.1%	10.3%	1.5%	0.2%
arXiv non-ML	181,570	**32.0%**	50.7%	13.7%	3.1%	0.6%
PubMed	1,679,954	**31.1%**	48.4%	16.4%	3.2%	0.9%
Barnett and Doubleday (2020)	1,112,345	30.0%	49.0%	n/a	n/a	n/a

Table 10: Acronym frequency distribution by venue (abstract acronyms). Each cell shows the percentage of unique acronyms appearing in that frequency range. The external baseline is Barnett and Doubleday (2020), PubMed titles and abstracts, 1950–2019. Single-use rates are comparable across venues. ML venues carry a larger 2–9 use bucket. Roughly 88 to 90 percent of ML acronyms appear fewer than ten times against 80 percent in PubMed. A value of n/a means that the source does not report the frequency range.

Acronyms	Share of distinct acronyms (%)	Share of occurrences (%)
Top 10	0.04	15.8
Top 20	0.09	21.4
Top 50	0.22	30.7
Top 100	0.45	38.4
Top 500	2.24	56.9

Table 11: Concentration of abstract acronym occurrences at the three conferences between 2015 and 2025. The corpus holds 22,332 distinct acronyms over 229,789 occurrences. A few hundred reused terms account for most of what a reader meets, while the 25.6 percent of acronyms that occur once carry only 2.5 percent of occurrences.

Metric	Better	Top 10%	Bottom 10%	Δ (top − bottom)
Flesch Reading Ease	↑ better	20.29 ± 0.34	18.49 ± 0.36	**+1.79**
Flesch–Kincaid Grade	↓ better	15.92 ± 0.07	16.44 ± 0.07	**-0.52**
Gunning Fog Index	↓ better	18.96 ± 0.08	19.52 ± 0.08	**-0.56**
SMOG Index	↓ better	16.72 ± 0.06	17.11 ± 0.06	**-0.39**
Dale–Chall Score	↓ better	13.09 ± 0.03	12.94 ± 0.03	+0.15
Spache Grade	↓ better	7.84 ± 0.02	7.98 ± 0.02	**-0.14**
Coleman–Liau Index	↓ better	16.78 ± 0.07	16.47 ± 0.07	+0.31
ARI	↓ better	18.10 ± 0.08	18.30 ± 0.08	**-0.20**
Linsear Write	↓ better	15.99 ± 0.11	16.40 ± 0.11	**-0.41**
LIX	↓ better	61.99 ± 0.20	62.67 ± 0.21	**-0.68**
RIX	↓ better	8.79 ± 0.06	9.11 ± 0.06	**-0.32**
FORCAST	↓ better	12.78 ± 0.02	12.71 ± 0.03	+0.07
Powers–Sumner–Kearl Grade	↓ better	8.34 ± 0.02	8.46 ± 0.02	**-0.12**
Words per sentence	↓ better	22.12 ± 0.14	23.19 ± 0.14	**-1.07**
Syllables per word	↓ better	1.94 ± 0.00	1.95 ± 0.00	-0.01

$n_{\text{top}} = 4,407$, $n_{\text{bottom}} = 4,285$ papers.

Table 12: NeurIPS, ICLR, and ICML readability for top-10 percent vs bottom-10 percent cited papers (within-year citation percentile, pooled through 2024). The most recent year is excluded because its citation counts are immature. Values are mean with 95 percent confidence interval (normal approximation). Bold indicates the gap favours the top-cited group by at least 0.05 in magnitude.

Prompt	Venue	Top 10%	Bottom 10%	Δ
Simple				
	NeurIPS	2.49 ± 0.48	2.19 ± 0.43	**+0.30**
	ICLR	2.36 ± 0.24	2.10 ± 0.23	**+0.25**
	ICML	2.29 ± 0.23	1.99 ± 0.22	**+0.30**
ASCB				
	NeurIPS	2.48 ± 0.81	2.20 ± 0.74	**+0.28**
	ICLR	2.26 ± 0.49	2.02 ± 0.44	**+0.24**
	ICML	2.19 ± 0.47	1.90 ± 0.42	**+0.29**
Own reasoning				
	NeurIPS	2.63 ± 0.57	2.32 ± 0.53	**+0.31**
	ICLR	2.55 ± 0.36	2.26 ± 0.34	**+0.28**
	ICML	2.46 ± 0.35	2.13 ± 0.32	**+0.33**

Table 13: LLM-as-judge readability for top-10% vs bottom-10% cited papers across NeurIPS, ICLR, and ICML (within-venue-year percentile, pooled through 2024). Each cell is the mean across the six models of that model's group-mean score (1–5, higher = easier), with a 95% CI over the six model means. Δ = top − bottom. Bold where top-cited papers are rated easier.

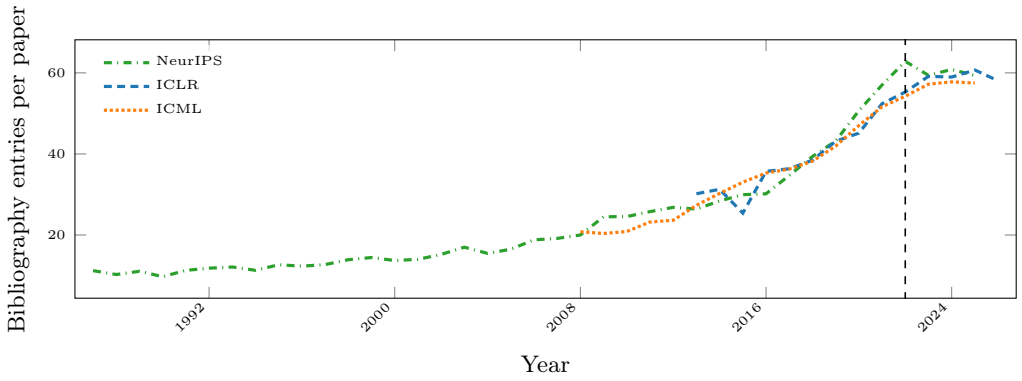


Figure 6: **Mean bibliography length per paper at the three conferences.** Mean number of references per paper by year at NeurIPS, ICLR and ICML, from Semantic Scholar reference counts. Papers without a reference list in Semantic Scholar are excluded.

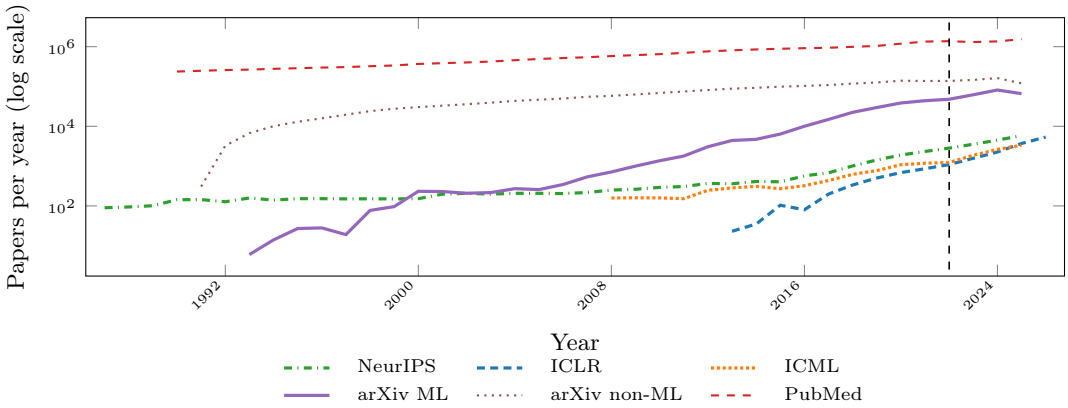


Figure 7: **Papers per year, 1987 to 2026.** arXiv ML submissions grew by roughly four orders of magnitude between 1993 and 2025. NeurIPS grew by a factor of roughly 65 between 1987 and 2025, from fewer than 100 accepted papers to approximately 5,800. PubMed and arXiv non-ML show sustained but slower growth over the same period. The volume increase is the structural backdrop against which the readability and acronym trends in Sections **??** and **??** must be interpreted.

Metric	Primary sources
Sentence length	**?** **?**]
Sign posting	**?** **?**]
Hedging	**?** **?** **?**]
Noun chunks	**?** **?** **?**]
Active narration	**?** **?** **?**]
Verbs	**?** **?**]
Nouns	**?**]
Numbers	**?**]
Sensational language	**?**]

Table 14: Writing style metrics audited per paper, with the original sources cited by **?**].

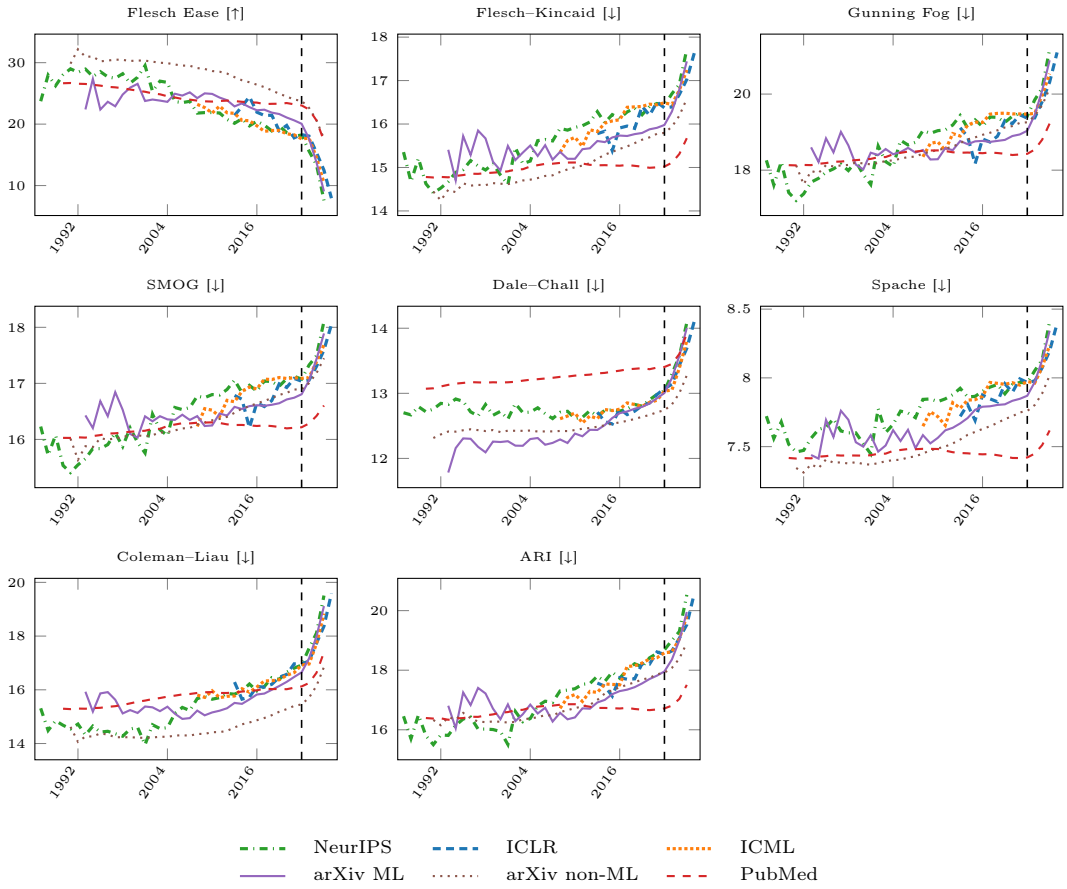


Figure 8: **Classical readability metrics, part A: Flesch family and grade-level formulas, 1987–2026.** Arrows indicate the direction of easier reading (↑ = higher is better, ↓ = lower is better). Every grade-level metric rises on NeurIPS and arXiv ML after 2015 and accelerates after 2020. Flesch Reading Ease drops correspondingly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part B with the remaining seven metrics appears in Figure 9.

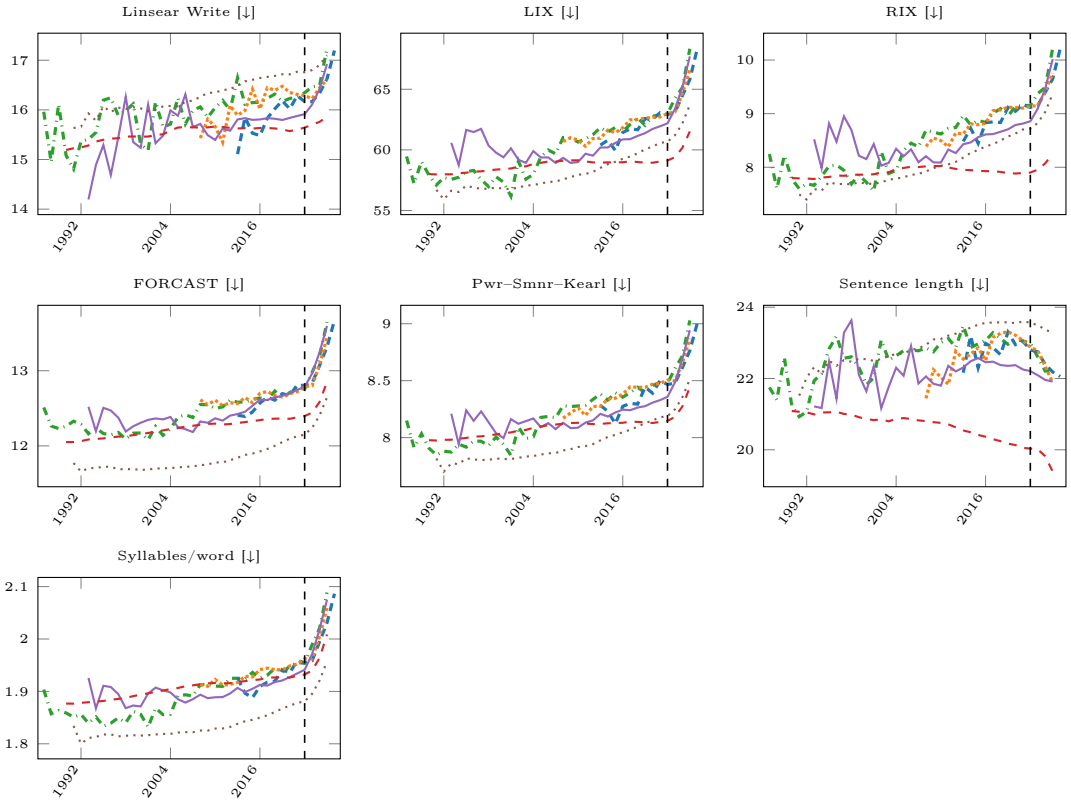


Figure 9: **Classical readability metrics, part B: Linsear Write, LIX, RIX, FORCAST, Powers–Sumner–Kearl, sentence length, and syllables per word, 1987–2026.** Arrows indicate the direction of easier reading (↓ = lower is better). Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part A with the Flesch family and the first eight grade-level formulas appears in Figure 8.

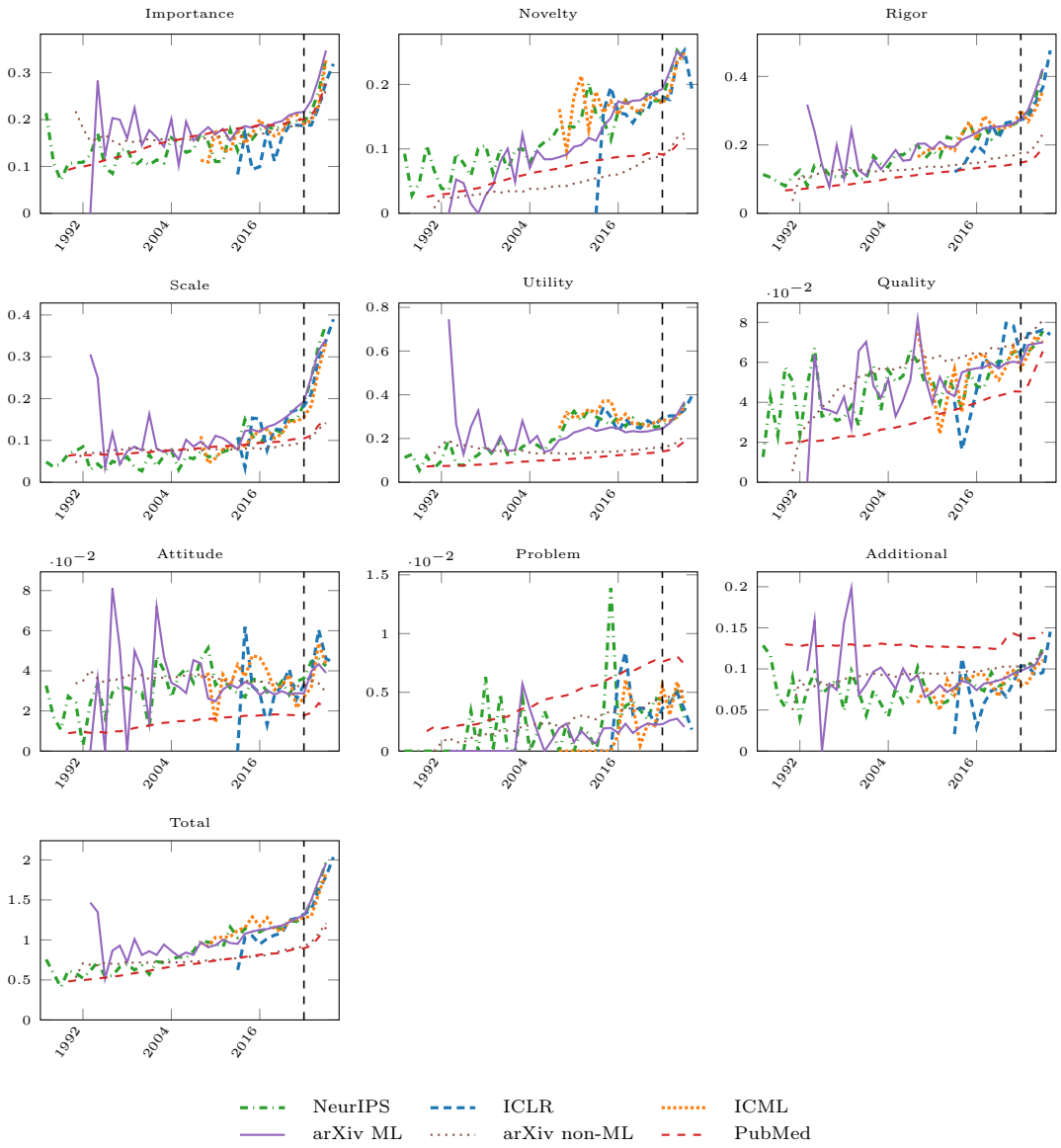


Figure 10: **All nine sensational language categories and their total per 100 abstract words, 1987–2026.** Novelty and scale rise sharply after 2015, reaching roughly 2.1 times and 3.4 times their 2015 levels on NeurIPS by 2025 (1.6 and 2.7 times on arXiv ML). The total rate and rigor rise more modestly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants.

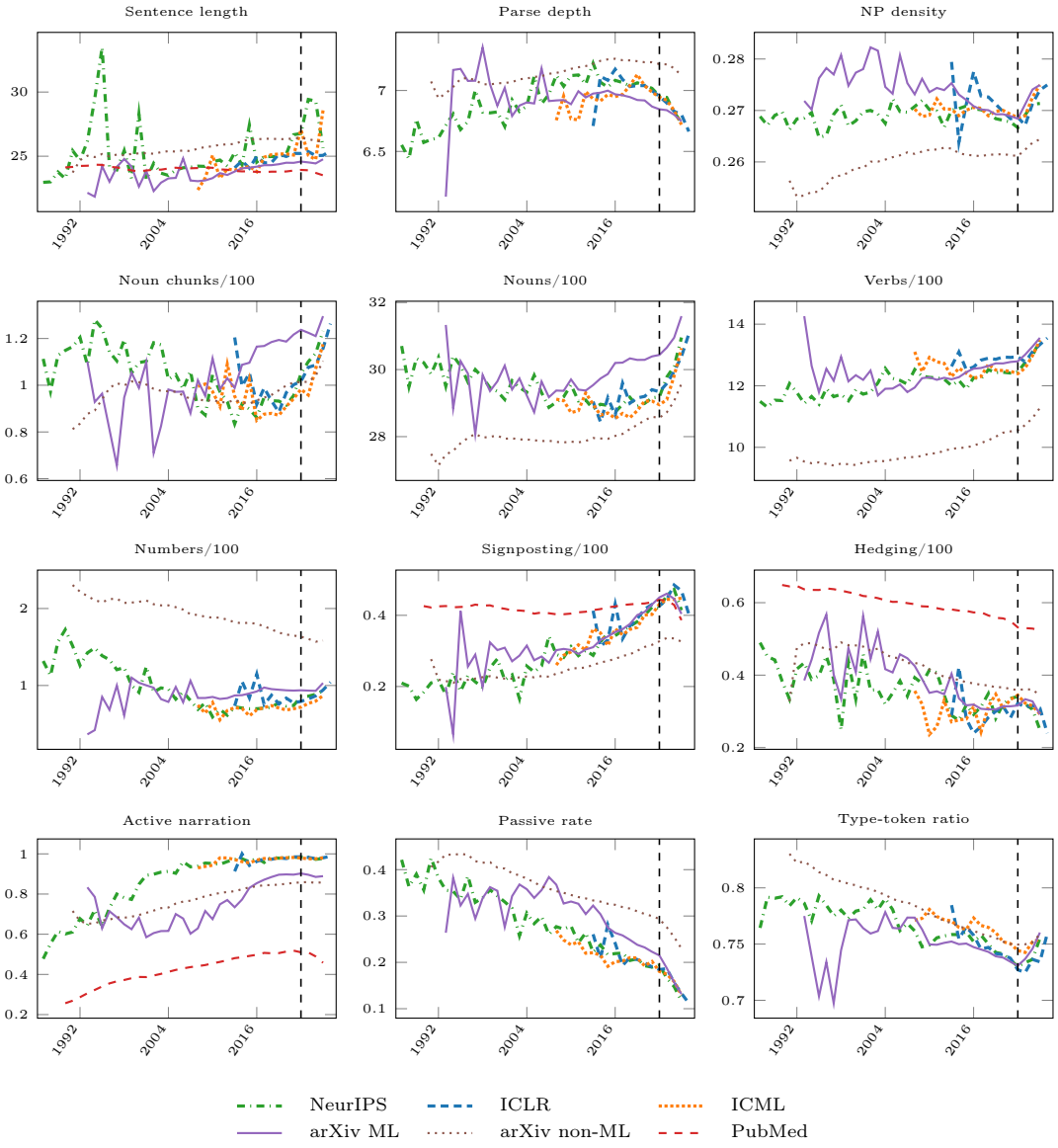


Figure 11: **Hohmann writing style metrics across all six corpora (NeurIPS, ICLR, ICML, arXiv ML, arXiv non-ML, and PubMed), 1987–2026.** PubMed data are available only for the four metrics computed by lightweight regex and word-level rules (sentence length, signposting, hedging, and active narration). The remaining metrics depend on full spaCy dependency parsing and were not run on the 24.3 million paper PubMed corpus because of compute cost. The corresponding panels omit PubMed. Across the suite, ML and non-ML venues are largely parallel.

Metric	A	Pct
Classical readability (15)		
Flesch Reading Ease	61.50	99.9%
Flesch–Kincaid Grade	7.80	0.0%
Gunning Fog	9.31	0.0%
SMOG	9.76	0.0%
Dale–Chall	12.73	31.3%
Spache	5.09	0.0%
Coleman–Liau	10.47	0.5%
ARI	8.97	0.1%
Linsear Write	7.43	0.2%
LIX	38.61	0.1%
RIX	3.27	0.1%
FORCAST	10.61	0.6%
Powers–Sumner–Kearl	5.91	0.1%
Words per sentence	12.54	0.3%
Syllables per word	1.57	0.3%
Acronyms		
Acronym density /100	5.12	89.0%
Novel acronyms (title)	0	n/a
Novel acronyms (abstract)	0	n/a

Metric	A	Pct
Writing style suite (12)		
Sentence length (Hohmann)	22.93	35.7%
Parse depth	5.47	6.7%
NP density	0.24	12.2%
Noun chunks /100	0.87	49.2%
Nouns /100	24.13	7.2%
Verbs /100	9.01	6.3%
Numbers /100	13.95	100.0%
Sign posting /100	0.00	24.3%
Hedging /100	0.58	77.1%
Active narration	1.00	52.2%
Passive rate	0.00	14.4%
Type–token ratio	0.66	13.3%
Sensational language (10)		
Importance	0.00	34.4%
Novelty	0.00	35.9%
Rigor	0.00	33.2%
Scale	0.00	35.3%
Utility	0.00	32.3%
Quality	0.00	44.7%
Attitude	0.00	46.7%
Problem	0.00	49.7%
Additional emphasis	0.00	42.3%
Total	0.00	8.6%

Table 15: Metric self-audit of this manuscript. Numbers are produced by running the same per-paper metric scripts used throughout the paper on this paper's own title and abstract. Percentile refers to the NeurIPS 1987–2025 abstract distribution for the same metric (n=30,595). Hohmann-suite and sensational-language metrics are computed at the abstract level only. The title falls below the readability scripts' minimum word count, so its classical scores are not reported. A percentile is not defined for a count of novel acronyms, which is shown as n/a. A = abstract, Pct = percentile.